

Utkarsha Tawalare

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ABOUT ME

Motivated Electronics and Telecommunication engineering student with hands-on experience in embedded systems and IoT-based project development. Skilled in microcontroller programming, sensor interfacing, and real-time system design through industrial and healthcare-focused projects such as automated weight batching, smart irrigation, and health monitoring systems. Passionate about building practical embedded solutions and seeking opportunities in core embedded and IoT roles.

EDUCATION

Vishwakarma Institute of Information Technology <i>Bachelor of Technology in Electronics and Telecommunication</i> CGPA: 8.20	Pune, Maharashtra 2024 – 2027
Government Polytechnic Amravati <i>Diploma in Electronics and Telecommunication</i> Percentage: 89.16%	Amravati, Maharashtra 2021 – 2024
Maharashtra State Board <i>SSC</i> Percentage: 86.00%	Maharashtra, India 2021

SKILLS

Programming Languages: C, C++
Microcontrollers: ESP, Arduino, 8051, LPC21xx, STM32
Tools & IDEs: STM32CubeIDE, Keil uVision, Arduino IDE, Proteus
Embedded Concepts: Timers, Interrupts, ADC, GPIO, UART, SPI, I2C, CAN, RTOS Basics
IoT Development: NodeMCU (ESP8266), Sensor Interfacing, Android App Integration
Soft Skills: Problem Solving, Teamwork, Communication
Languages: English, Hindi, Marathi

EXPERIENCE

MDB Electrosoft Pvt. Ltd. <i>Embedded & IoT Intern</i> - Completed a 6-month intensive internship focused on Embedded Systems and IoT development . - Programmed microcontrollers including Arduino and NodeMCU (ESP8266) for various applications. - Executed hardware-software integration through sensor interfacing and basic PCB design . - Developed Android application integrations to monitor and control IoT systems remotely.	On-site Dec 2023 – May 2024
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PROJECTS

Industrial UPS Weight Batching System <i>Microcontroller — Load Sensors — Real-time Control</i> Developed an automated weight batching system utilizing load sensors for high-precision industrial UPS applications .	On-going Industrial Project
IoT Smart Irrigation System <i>NodeMCU — WiFi — Soil Sensors</i> Built a WiFi-enabled smart irrigation system for automated water management using soil moisture sensors and NodeMCU .	IoT Project
IoT Health Monitoring System <i>Biomedical Sensors — Cloud Dashboard</i> Designed a wireless patient monitoring system that tracks vital signs via biomedical sensors and displays data on a cloud-based dashboard .	Healthcare Project
Assistive Smart Glove for Paralysis Patient <i>Flex Sensors — Gesture Detection</i> Created a wearable assistive device using flex sensors to detect hand gestures, aiding in communication and rehabilitation for paralysis patients .	Wearable Tech
Multi-Mode Robotics Platform <i>Bluetooth — Obstacle Avoidance — Line-Following</i> Developed an autonomous robotics platform with integrated Bluetooth control and multiple operational modes including line-following and obstacle avoidance .	Robotics